

DFX LOG

BLDC Motor Efficiency Testbench - 3D Printing Design Rationale

Overview

This log documents design-for-manufacture (DFM), design-for-assembly (DFA), and design-for-cost (DFC) decisions for every 3D printed part in the testbench, alongside structural verification results.

Part: Motor Reaction-Torque Arm

Name	Motor Reaction-Torque Arm
Function	Mounts the A2212 motor to a pivoting cradle. Motor reaction torque swings the arm against a fixed load cell, forming the testbench's reaction-torque dynamometer.
Manufacturing	FDM 3D printing
Material	Generic PLA
Status	CAD complete. FEA (torsion) ran in FreeCAD.
Phase	Design Phase

1. Design Decisions and DFX Rationale

Design feature	DFX Rationale
Flat mounting tab with countersunk hole	- Provides a flush connection to the load cell arm, keeping the fastener head from interfering with adjacent geometry. - Flush connections allow for easier transportation without scratching surrounding objects (work-in-progress stacking).
Diagonal gusset (rather than a right-angle L-bracket)	- Resists bending and torsional deflection far better than an unsupported right-angle bracket of the same material and print process. - Triangulates the load path between the mounting tab and the motor boss.
Cylindrical motor boss with 4-hole bolt circle and center bore	- Matches the A2212 motor's standard mounting pattern directly, so the part bolts straight to the motor face without the use of an adapter. - Cylindrical shape allows maneuverability of the motor wires to attach to the ESC (Electronic Speed Controller) without designing a specific guide for the wires.
Fillets at gusset-to-plate and gusset-to-boss junctions	Reduces stress concentration at internal corners (necessary for Fused Deposition Modelling (FDM) PLA), which is more notch-sensitive and brittle than machined metal.

Print orientation: layers aligned along the gusset's long axis	FDM parts are anisotropic and the reaction torque loads the gusset in torsion; orienting the print so layer lines run along the load path keeps the load within continuous layers instead of shearing across weak interlayer bonds.
--	---

2. Structural Verification (FEA)

Torsion analysis performed in FreeCAD's FEM workbench, treating the motor's reaction torque as the applied load and generic PLA as the assigned material.

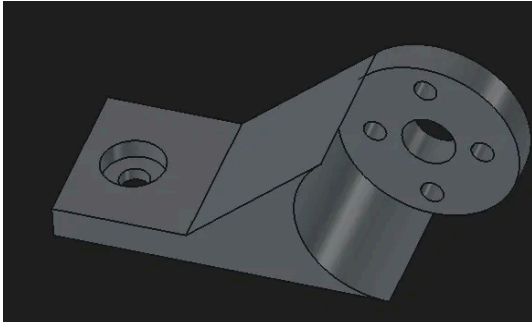
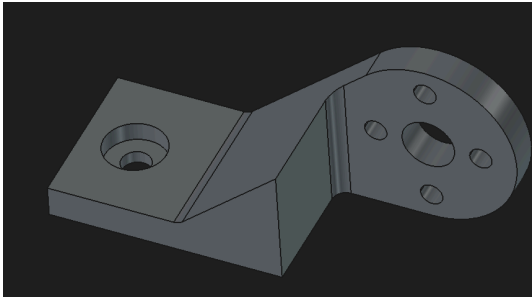
2.1 Material Inputs

Property	Value Used
Software	FreeCAD 1.1.0
Material	Generic PLA (FreeCAD material library)
Density	1240 kg/m ³
Young's Modulus	3640 MPa
Poisson's ratio	0.36
Yield strength (<i>library value</i>)	35.9 MPa
Ultimate tensile strength (<i>corrected</i>)	40 MPa <i>Note: original library value of 26.4 MPa was below yield strength, which violates the definition Ultimate Tensile Strength ≥ Yield Strength. Therefore, corrected to 40 MPa, consistent with published bulk PLA data range and not greatly exceeding library value for yield strength for conservative estimations.</i> <i>Source:</i> 32.938 MPa - https://3dprint.com/233916/tensile-properties-of-3d-printed-pla/ 60 MPa - https://pmc.ncbi.nlm.nih.gov/articles/PMC10880662/#s009
Thermal properties	Not used <i>As no thermal load is applied in this test, will revisit if future test models heat soak from the motor into the mount during extended runs.</i>

2.2 Design History

Iteration	Max Stress	Max Displacement	Safety Factor	Corrected Safety Factor*
1 (original)	6.18 MPa	0.15 mm	~6.47	~3.06
2 (current)	4.89 MPa	0.17 mm	~8.18	~3.86

* There can be a 52.8% decrease in the strength of the material depending on if there is a mistake in orientation. Therefore a corrected safety factor is also being considered in this design. Source: E A Syaefudin et al 2023 J. Phys.: Conf. Ser. 2596 012002

Iteration	CAD View	Change Made
1 (original)	 Original Design	To allow for motor housing support the original gusset-to-boss junction incorporated a compound curvature transition (concave-to-convex in immediate succession) at the boundary. However, such curvature inflections constitute a geometric discontinuity and elevate the local stress concentration factor independent of applied load magnitude in linear elastic range (Peter's Concentration Stress Factor).
2 (current)	 Current Design	This led to a complete geometry redesign . The compound curvature was the root cause of the fillet and chamfer generation failures encountered in FreeCAD, due to the limitation of the software, blend operations require a well-defined, continuous curvature along the target edge. To simplify the junction, a single flat-wall transition was then successfully filleted. The reduction in peak stress and displacement is attributed by the reduction of geometric discontinuity, and the stress-smoothing effect of the fillet at the corners.

2.3 Criteria for FEA

Input	Details
Gravitation Constraint (Self Weight)	The whole body had a gravity component to it as the part is to be suspended
Fixed Constraint	At the countersunk hole on the flat tab (bolted connection to the load cell arm)
Torque Load (4x Force Load) <i>Note: since FreeCAD does not have a native torque load mode, 4 force load were applied to create a moment.</i>	4 N applied on each force load. Radius from motor mount holes to centre of motor: 9.5mm 3s LiPo - Peak Voltage: 12.6 V (during bust) Propeller 1045, current draw ~ 16.0 A* A2212 BLDC 1000 kV Torque Constant: $\frac{60}{2\pi} \div 1000 \text{ kV} = 0.00955 \text{ N}\cdot\text{m/A}$ Torque (τ): $0.00955 \times 16 \approx 0.153 \text{ N}\cdot\text{m}$ Force: $\tau/r = 0.153 \text{ N}\cdot\text{m} / 9.5 \text{ mm} = 16.08 \text{ N}$ (or 4 N at each motor mount hole)
Force Load (Motor Weight)	51g A2212 BLDC motor $\times 9.81 \text{ m/s}^2 = 500 \text{ mN}$
Force Load (Propeller Thrust)	800 approx. grams of thrust = 8 N Source - technical description from supplier: https://quartzcomponents.com/products/a2212-1000kv-blDC-drone-motor

2.4 Results for Current Design

Metric	Result
Max von Mises stress	4.89 MPa
Max displacement	0.17 mm
Safety factor (bulk material)	~8.18 (40 MPa / 4.89 MPa)
Interpretation	Excellent margin even before accounting for print-orientation strength loss. At bulk-material SF ~8.18, even the weaker cross-layer FDM direction (50–70% of bulk) gives an effective SF of roughly 3.3 - 5.7. Correct print orientation will be practiced but is no longer the only margin of safety.
Action taken	Reduced geometric discontinuity, added fillets at stress loaded boundaries. Reduced printing material without jeopardizing structural integrity.

3. Next Iteration / Steps

- Load case confirmed: gravity constraint applies self-weight to the whole bracket (not the motor's mass), combined with the motor's reaction torque. The fixed constraint is the bolted face at the countersunk hole.
- Double-check the FreeCAD displacement plot's colour legend before relying on it visually; it displayed a symmetric $\pm 0.5\text{mm}$ range for a magnitude result, which should never be negative. Confirm the reported 0.17mm figure comes from the results table rather than the plot, and re-scale the legend if revisiting this view.
- *16A current draw is based on 33.3% headroom on max current draw of 12A for 60s. This will be revisited once an INA226 is available to test the true burst current draw of the motor.
- Safety factor (~ 8.18 , bulk material) is now comfortable even accounting for FDM interlayer strength loss. Verify wall thickness between the center bore and the four bolt holes on the motor boss is still sufficient given bolt preload.
- Record final print orientation and slicer settings (layer height, infill %, wall count) once printing begins as these affect the real-world strength independent of the CAD geometry.